

Science as a human endeavour

Verbal/Linguistic

Many scientists contributed to our understanding of the elements, their properties and the family relationships between them. Each of these scientists therefore contributed to the development of the periodic table we use today. Some of these scientists are discussed in the Science as a Human Endeavour on pages 115–16 of your student book. This table provides some more information about them.

Scientist	Lived	Scientific contributions	Other facts
Antoine-Laurent de Lavoisier	1743–93	• Did not discover any new elements • Identified and named the existing 33 elements • Proved that mass was not destroyed in a reaction • Proved that rusting involved oxygen bonding to iron (increasing its mass) • Proved that combustion needed oxygen	• Nationality: French • Was very wealthy; worked as a tax collector • Had 13 000 beakers in his laboratory • Was executed by guillotine after the French Revolution • His statue in Paris was made with a 'spare' head from another statue
John Dalton	1766–1844	• Revived the Ancient Greek idea of atoms • Gave each element a pictorial symbol, which is not used today • Predicted atomic masses • Developed laws that explain how gases behave • Showed that compounds were made of elements in definite mass proportions	• Nationality: English • Came from a very poor family • Was put in charge of his small school when he was 12 years old • Still taught mathematics at a primary school when he was old and famous
Jons Jacob Berzelius	1779–1848	• Used atomic mass to arrange the elements • Gave the elements symbols that we still use today, based on their Greek or Latin names • Showed that compounds were made of elements in definite mass proportions • Was the first to classify substances as organic (containing C) or inorganic (no C)	• Nationality: Swedish • Used superscript numbers instead of subscript numbers in chemical formulas—wrote water as H^2O instead of H_2O
Johann Dobereiner	1780–1849	• Placed elements into 'triads' (families of three)	• Nationality: German

Scientist	Lived	Scientific contributions	Other facts
John Newlands	1837–98	- Used atomic mass to arrange elements - Noticed every eighth element 'harmonising' (forming patterns) - Placed elements in periods of 7 elements	- Nationality: English - Called his arrangement 'the law of octaves' after musical octaves
Lothar Meyer	1830–95	- Constructed a table very similar to Mendeleev's - Finished his periodic table before Mendeleev - Published his table after Mendeleev	- Nationality: German - Is regarded as a 'father' of the periodic table
Dmitri Ivanovich Mendeleev	1834–1907	- Used atomic mass to arrange elements - Like Newlands, arranged the elements in periods of 7 - Arranged related elements in columns like Dobereiner's triads - Left gaps for undiscovered elements (e.g. Ge)	- Nationality: Russian - Refused to accept that electrons and radiation and radioactivity existed - Has a radioactive element named after him—mendelevium!
William Ramsay	1852–1916	- Discovered the noble, or inert, gases - Set them as a separate group	- Nationality: Scottish - Received the Nobel Prize in Chemistry
Henry Moseley	1887–1915	- Used atomic number, not mass, to arrange elements - Produced the periodic table still in use today	- Nationality: English - Was shot through the head and killed in World War I at Gallipoli

harmonise (v) to be similar in some way

octave (n) a group of eight notes in music

Use the information in the table and information from Unit 4.2 in your student book to **identify**:

(a) the scientist who was Scottish _____

(b) the scientist who finished his table before Mendeleev did but received less recognition _____

(c) the scientist who thought that the elements harmonised in octaves like music _____

(d) the scientist who received a Nobel Prize _____

- (e) the scientist who wrote water as H_2O _____
- (f) the scientist who left gaps in his table _____
- (g) the scientist who developed triads _____
- (h) the scientist who thought that light was an element _____
- (i) the scientist who is considered to be the 'father' of the modern periodic table

- (j) the scientist who developed gas laws _____
- (k) the scientist who arranged elements according to their atomic number and not their mass _____
- (l) the scientist who was shot in the head _____
- (m) the scientist who had his head chopped off _____
- (n) the scientist who was Russian _____
- (o) the scientist who revived the idea that atoms existed _____
- (p) the scientist who had 13 000 beakers in his laboratory _____
- (q) the scientist who was put in charge of his school when he was 12

- (r) the scientist who grouped chemicals as organic or inorganic _____
- (s) the scientist who gave the elements the symbols we use today _____
- (t) the scientist who has a radioactive element named after him, even though he didn't believe in radioactivity _____
- (u) the scientist who proved that iron got heavier when it rusted _____
- (v) the scientist who had a statue that should have had its head chopped off because it was the wrong one! _____